

# Symbolic Projection Notes for $B_{c1} \rightarrow B_c^{(*)} \gamma$

Working notes

July 11, 2026

## 1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \rightarrow B_c \gamma, \quad B_{c1}(6743) \rightarrow B_c^* \gamma, \quad B_{c1}(6750) \rightarrow B_c \gamma, \quad B_{c1}(6750) \rightarrow B_c^* \gamma.$$

The calculation is structurally different from the  $D_s$  and  $B_s$  light-cone sum rules. In the  $B_c$  system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no  $\chi_s \langle \bar{s}s \rangle$  type term.

## 2 Currents and Propagators

For  $B_c^+ \sim c\bar{b}$ , we use

$$j_5 = \bar{b} i \gamma_5 c, \quad j_\rho^V = \bar{b} \gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} P^\alpha \gamma_5 c, \quad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B.$$

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \quad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q) i e_Q \gamma_\nu \epsilon^\nu S_Q^{(0)}(k).$$

For the  $\bar{b}$  line it is convenient to use the effective antiquark charge  $e_{\bar{b}} = +e/3$ , while  $e_c = +2e/3$ . Equivalently, one may keep  $e_b = -e/3$  and track the orientation of the antiquark line. The scripts use  $e_{\bar{b}}$  explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a  $B_c$ -mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted  $B_c$ -mixing work uses the same local basis,

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} p^\alpha \gamma_5 c,$$

with  $m_b = 4.18$  GeV,  $m_c = 1.27$  GeV, and the final working window

$$7 \text{ GeV}^2 \leq M^2 \leq 9 \text{ GeV}^2, \quad 53 \text{ GeV}^2 \leq s_0 \leq 55 \text{ GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

### 3 $B_{c1} \rightarrow B_c \gamma$ : E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q) g_{\mu\nu}, \quad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p \cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu} \mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \quad p \cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$\begin{aligned} C_A^{(c)} &= -4i m_c + \frac{4i m_b m_c^2}{pq} - \frac{4i m_c^3}{pq}, \\ C_A^{(\bar{b})} &= -4i m_b + \frac{4i m_b^3}{pq} - \frac{4i m_b^2 m_c}{pq}, \end{aligned}$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar{b}} C_A^{(\bar{b})}.$$

For the tensor current,

$$\begin{aligned} C_B^{(c)} &= \frac{4i m_b m_c - 8i m_c^2}{m_b + m_c} - \frac{4i m_c^2 p_2}{(m_b + m_c) pq}, \\ C_B^{(\bar{b})} &= -\frac{4i m_b m_c}{m_b + m_c} - \frac{4i m_b^2 p_2}{(m_b + m_c) pq}, \end{aligned}$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar{b}} C_B^{(\bar{b})}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

### 4 $B_{c1} \rightarrow B_c^* \gamma$ : Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q) \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for  $q^2 = 0$ . The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p_2(pq)^2, \quad V_{\mu\nu\rho} \mathcal{P}^{\mu\nu\rho} = 1.$$

The projected  $J_A$  triangle cores are

$$C_{A,V}^{(c)} = -\frac{4i m_b m_c}{p_2} + \frac{2i m_c^2}{p_2} + \frac{4i m_c^2}{pq},$$

$$C_{A,V}^{(\bar{b})} = -\frac{2i m_b^2}{p_2} + \frac{4i m_b m_c}{p_2} + \frac{4i m_b^2}{pq}.$$

The tensor-current vector cores are

$$C_{B,V}^{(c)} = \frac{2i m_c}{m_b + m_c} + \frac{2i m_b^2 m_c - 4i m_b m_c^2 + 2i m_c^3}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b m_c^2 - 4i m_c^3}{(m_b + m_c)pq} + \frac{2i m_c pq}{(m_b + m_c)p_2},$$

$$C_{B,V}^{(\bar{b})} = \frac{2i m_b}{m_b + m_c} + \frac{-6i m_b^3 + 4i m_b^2 m_c + 2i m_b m_c^2}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b^3 - 4i m_b^2 m_c}{(m_b + m_c)pq} + \frac{2i m_b pq}{(m_b + m_c)p_2}.$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar{b}} C_{A,V}^{(\bar{b})}, \quad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar{b}} C_{B,V}^{(\bar{b})}.$$

## 5 First Numerical Hard-Photon Pass

The script `Bc_gamma/scripts/bc_radiative_pilot_sumrule.py` uses the compact triangle cores above to produce a first hard-photon numerical estimate. For this pass we use the same central heavy-quark inputs and Borel window as the submitted  $B_c$ -mixing analysis,

$$m_b = 4.18 \text{ GeV}, \quad m_c = 1.27 \text{ GeV}, \quad \theta = 43.3^\circ,$$

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \quad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The decay-constant inputs are

$$f_{B_c} = 0.371 \text{ GeV}, \quad f_{B_c^*} = 0.384 \text{ GeV}, \quad f_{B_{c1}} = 0.373 \text{ GeV}.$$

The first two are standard external benchmark inputs. The last is a common axial-vector benchmark used only for this first pass; separate physical  $f_1$  and  $f_2$  residues have not yet been extracted from a two-point mixed-current  $B_c$  sum rule in this folder.

Table 1: Pilot hard-photon results in the  $B_c$  sector. The intervals show the envelope over  $M^2 = \{7, 8, 9\} \text{ GeV}^2$  and  $s_0 = \{53, 54, 55\} \text{ GeV}^2$ .

| Channel                                 | Status                | Coupling/projection coefficient             | $\Gamma$ (keV)                  |
|-----------------------------------------|-----------------------|---------------------------------------------|---------------------------------|
| $B_{c1}(6743) \rightarrow B_c \gamma$   | hard-QCDSR baseline   | $+0.0219 [0.0209, 0.0232] \text{ GeV}^{-1}$ | $0.108 [0.098, 0.121]$          |
| $B_{c1}(6750) \rightarrow B_c \gamma$   | hard-QCDSR baseline   | $-0.354 [-0.387, -0.331] \text{ GeV}^{-1}$  | $29.6 [25.9, 35.3]$             |
| $B_{c1}(6743) \rightarrow B_c^* \gamma$ | diagonal vector pilot | $-0.0620 [-0.0669, -0.0583]$                | $46.3 [41.0, 53.8]$             |
| $B_{c1}(6750) \rightarrow B_c^* \gamma$ | diagonal vector pilot | $+0.362 [0.335, 0.401]$                     | $1.67 [1.43, 2.04] \times 10^3$ |

The  $B_c \gamma$  entries are the controlled hard-photon baseline from the present calculation. The  $B_c^* \gamma$  entries should be read as diagnostic numbers only. They are obtained from the gauge-invariant  $V_{\mu\nu\rho}$  projection and a diagonal single-variable reduction. Before using them as final paper results one must match this coefficient to the physical  $1^+ \rightarrow 1^- \gamma$  tensor basis and carry out the full double-Borel analysis, including denominator-cancellation or contact terms.

## 6 Next Step

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \quad pq = \frac{P^2 - p^2}{2}.$$

For  $B_c\gamma$ , the structure is close to the strange-sector E1 case, but with two heavy masses. For  $B_c^*\gamma$ , the additional  $1/p_2$  and  $pq/p_2$  structures mean that the double dispersion should be written explicitly before numerical Borel analysis. A diagonal single-variable replacement can be useful as a calibration check, but it should not be the final derivation for the vector final state.

### Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2.$$

With Feynman parameters  $x, y, z$ ,  $x + y + z = 1$ , the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1 - z)m_c^2 + zm_b^2 - zyp^2 - xzP^2.$$

For photon emission from the  $\bar{b}$  line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2,$$

and the analogous reduction gives, with  $x + y + z = 1$ ,

$$\Delta_{\bar{b}} = xm_c^2 + (1 - x)m_b^2 - x(1 - x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in  $p^2$  and  $P^2$ , not simply the two-point mixing correlator with an extra charge factor.